

Wireshark Guide

filters · tips · tricks · tshark

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swipe to start »

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PROTOCOLS

decoded out of the box

USERS

analysts and engineers

OPEN

source · free forever

01

Troubleshoot latency, packet loss and routing

02

Investigate incidents and PCAP forensics

03

Intercept credentials and map protocols

04

Detect C2, DNS tunneling and anomalies

◆ **IMPORTANT**

 @kerem.tech

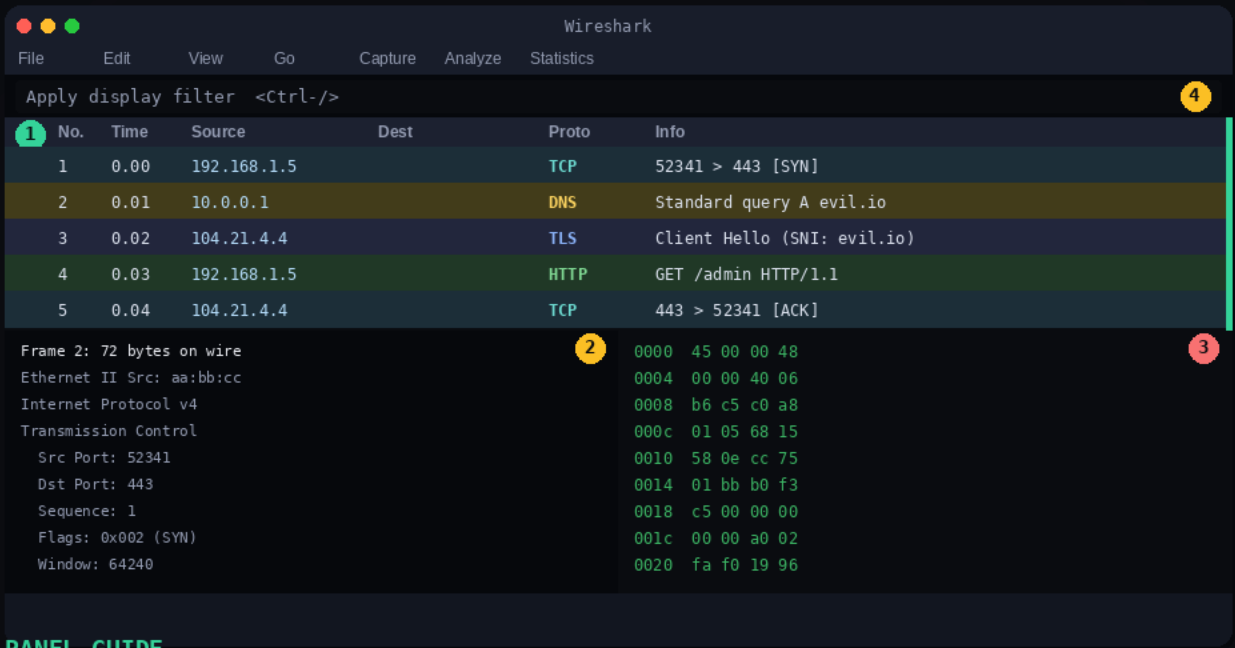


X @keraattin

The UI Explained

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Four areas · one workflow · total packet visibility.



PANEL GUIDE

- 1 Packet List** · All captured packets · click one to inspect details
- 2 Packet Details** · Expandable protocol tree of the selected packet
- 3 Packet Bytes** · Raw hex and ASCII dump of the packet
- 4 Filter Bar** · Type your display filter here · green = valid

♦ PRO TIP

Drag column borders to resize · right-click to add columns.
F11 hides the menu and toolbar for distraction-free analysis.

Capture vs Display

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Two completely different filter systems.

PRE-CAPTURE

Capture Filters

VS

POST-CAPTURE

Display Filters

Before capture

WHEN APPLIED



After capture, anytime

SYNTAX

Berkeley Packet Filter



Wireshark syntax

LIVE CHANGES

No · restart needed



Yes · instant

EFFECT

Packets not stored



Packets hidden, not lost

EXAMPLE

host 10.0.0.1



ip.addr==10.0.0.1

EXAMPLE

port 443



tcp.port==443

♦ RULE OF THUMB

Use capture filters only when storage is critical.
Use display filters by default · safer and faster.

♦ WHEN TO USE CAPTURE FILTERS

Multi-hour captures on production links.
Storage-constrained devices and SPAN ports.
When you know exactly what protocol to grab.

Display Filter Syntax

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Learn these rules once · use them in every capture.

COMPARISON OPERATORS

==

equals

`ip.addr == 10.0.0.1`

!=

not equal

`tcp.port != 80`

contains

string contains

`http.host contains "corp"`

matches

regex match

`dns.qry.name matches "\.ru$"`

gt / lt

greater · less

`frame.len gt 1400`

LOGICAL OPERATORS

&&

both must match

`ip.addr==10.0.0.1 && tcp.port==443`

||

either matches

`tcp.port==80 || tcp.port==8080`

!

exclude matches

`!arp && !icmp`

♦ QUICK REFERENCE

Green bar = valid filter · Red bar = syntax error

Right-click any field > Apply as Filter

Noise Elimination

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Clear the capture before you start looking.

Wireshark · capture.pcap

File Edit View Go Capture Analyze Statistics Telephony Tools

Filter: !(arp || icmp || ssdp || mdns || llmnr || nbns)

No.	Time	Source	Destination	Protocol	Length	Info
14	0.412031	192.168.1.5	8.8.8.8	DNS	72	Standard query A example.com
21	1.003812	192.168.1.5	104.21.88.14	TCP	74	443 > 52341 [SYN] Seq=0
35	1.887204	104.21.88.14	192.168.1.5	HTTP	512	HTTP/1.1 200 OK
48	2.341009	192.168.1.23	192.168.1.5	TCP	1460	8080 [PSH, ACK]
61	3.112887	192.168.1.5	10.0.0.1	TLS	517	Client Hello

Filter active · ARP, ICMP, SSDP, mDNS, LLMNR, NBNS hidden

!(...) ▶ Negate everything inside · exclude these protocols

arp ▶ Link-layer broadcasts · constant background noise

icmp ▶ Ping and traceroute · floods view but rarely matters

ssdp ▶ UPnP IoT chatter · not useful for investigation

mdns ▶ Bonjour local resolution · zero investigation value

llmnr ▶ Windows LAN name resolution · pure noise

nbns ▶ Legacy NetBIOS broadcasts · pre-DNS noise

Busy networks flood you with ARP, mDNS and NetBIOS.
This filter removes all six noise sources at once,
leaving only routed traffic worth investigating.

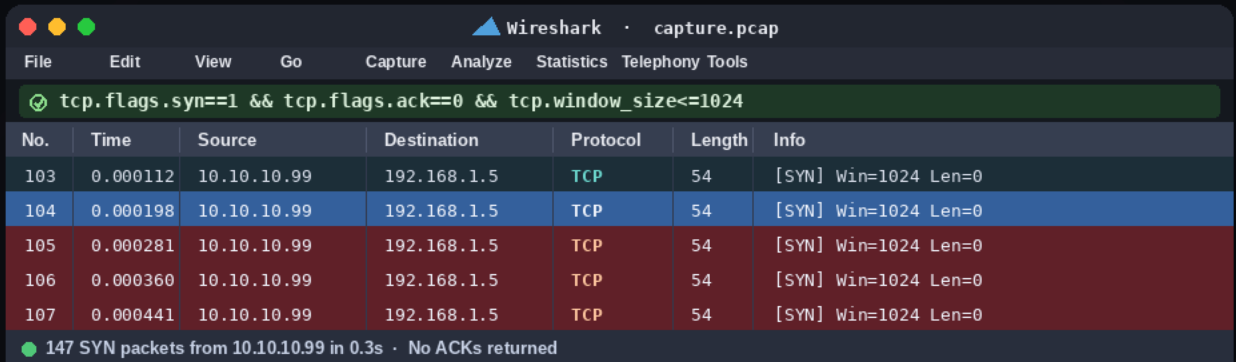
♦ WHEN TO USE THIS

First thing to apply on any noisy capture file.
Triage step before deeper protocol analysis.

Port Scan Detection

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Identify scanners by their TCP fingerprint.



A screenshot of the Wireshark network protocol analyzer interface. The title bar shows 'Wireshark · capture.pcap'. The menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, and Telephony Tools. A filter bar at the top displays the expression 'tcp.flags.syn==1 && tcp.flags.ack==0 && tcp.window_size<=1024'. Below this is a packet list table with columns: No., Time, Source, Destination, Protocol, Length, and Info. The table shows five TCP SYN packets from 10.10.10.99 to 192.168.1.5, all with window size 0. A status bar at the bottom indicates '147 SYN packets from 10.10.10.99 in 0.3s · No ACKs returned'.

No.	Time	Source	Destination	Protocol	Length	Info
103	0.000112	10.10.10.99	192.168.1.5	TCP	54	[SYN] Win=1024 Len=0
104	0.000198	10.10.10.99	192.168.1.5	TCP	54	[SYN] Win=1024 Len=0
105	0.000281	10.10.10.99	192.168.1.5	TCP	54	[SYN] Win=1024 Len=0
106	0.000360	10.10.10.99	192.168.1.5	TCP	54	[SYN] Win=1024 Len=0
107	0.000441	10.10.10.99	192.168.1.5	TCP	54	[SYN] Win=1024 Len=0

147 SYN packets from 10.10.10.99 in 0.3s · No ACKs returned

tcp.flags.syn==1 ▶ SYN flag set · connection initiation packet

tcp.flags.ack==0 ▶ ACK not set · this is not a SYN-ACK reply

tcp.window_size<=1024 ▶ Tiny window · classic scanner fingerprint

&& ▶ All three conditions must match at once

Legitimate clients open a few connections with large windows.
Scanners fire hundreds of SYNs per second with window sizes
of 1024 or less · this filter makes the scanner's IP obvious.

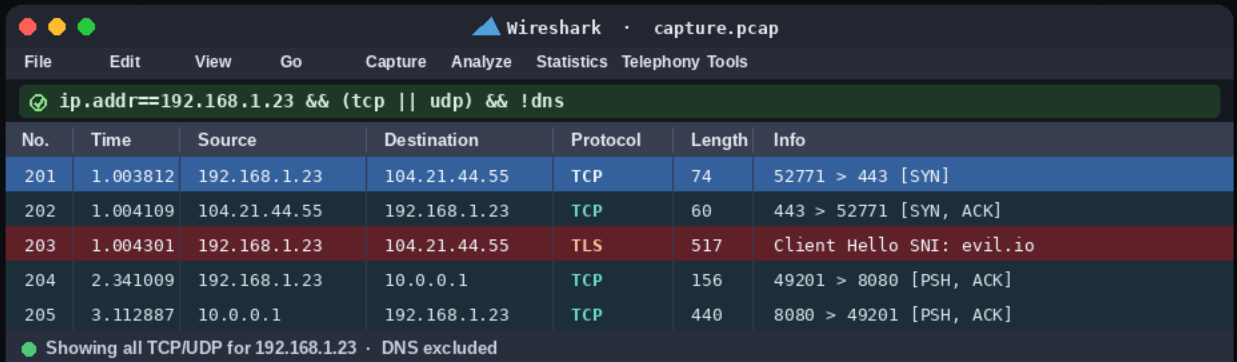
♦ WHEN TO USE THIS

When you suspect reconnaissance activity in a capture.
Useful for triaging IDS alerts and noisy hosts.

Host Isolation

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Lock onto one IP · follow every move it makes.



Wireshark · capture.pcap

File Edit View Go Capture Analyze Statistics Telephony Tools

ip.addr==192.168.1.23 && (tcp || udp) && !dns

No.	Time	Source	Destination	Protocol	Length	Info
201	1.003812	192.168.1.23	104.21.44.55	TCP	74	52771 > 443 [SYN]
202	1.004109	104.21.44.55	192.168.1.23	TCP	60	443 > 52771 [SYN, ACK]
203	1.004301	192.168.1.23	104.21.44.55	TLS	517	Client Hello SNI: evil.io
204	2.341009	192.168.1.23	10.0.0.1	TCP	156	49201 > 8080 [PSH, ACK]
205	3.112887	10.0.0.1	192.168.1.23	TCP	440	8080 > 49201 [PSH, ACK]

Showing all TCP/UDP for 192.168.1.23 · DNS excluded

ip.addr==X.X.X.X ▶ Match any packet where this IP is src or dst

(tcp || udp) ▶ Only transport layer traffic · no raw IP noise

!dns ▶ Exclude DNS · focus on real application sessions

&& ▶ All conditions must be true simultaneously

When a host looks suspicious, remove everything else.

This filter shows all TCP and UDP sessions for that IP while stripping DNS clutter from the view.

◆ WHEN TO USE THIS

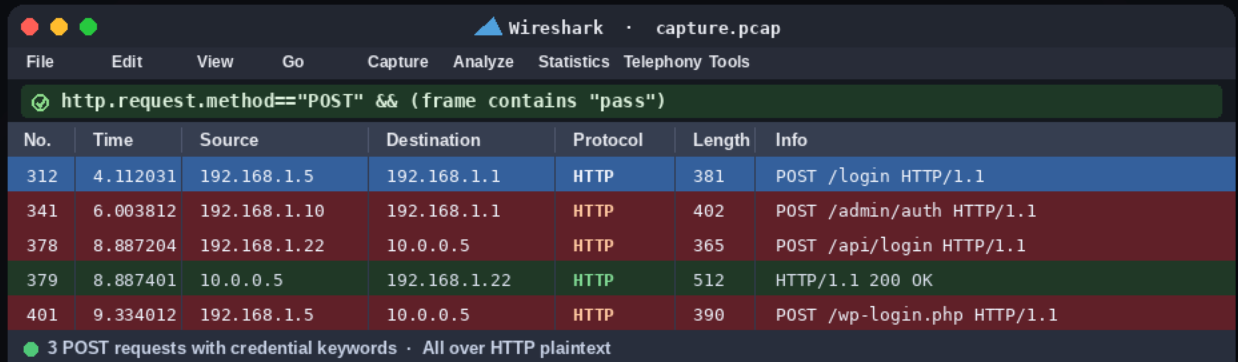
After identifying a suspect host from logs or alerts.

When tracing an attacker's movement across sessions.

Credential Hunt

Find plaintext credentials crossing the wire.

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Wireshark · capture.pcap

File Edit View Go Capture Analyze Statistics Telephony Tools

✓ http.request.method=="POST" && (frame contains "pass")

No.	Time	Source	Destination	Protocol	Length	Info
312	4.112031	192.168.1.5	192.168.1.1	HTTP	381	POST /login HTTP/1.1
341	6.003812	192.168.1.10	192.168.1.1	HTTP	402	POST /admin/auth HTTP/1.1
378	8.887204	192.168.1.22	10.0.0.5	HTTP	365	POST /api/login HTTP/1.1
379	8.887401	10.0.0.5	192.168.1.22	HTTP	512	HTTP/1.1 200 OK
401	9.334012	192.168.1.5	10.0.0.5	HTTP	390	POST /wp-login.php HTTP/1.1

3 POST requests with credential keywords · All over HTTP plaintext

`http.request.method=="POST"` ▶ Only POST requests · where form data is submitted

`frame contains "pass"` ▶ Raw payload contains the string "pass"

`frame contains "user"` ▶ Or contains "user" anywhere in the frame

`||` ▶ Any keyword match is enough to flag the packet

HTTP form submissions travel in cleartext. This filter surfaces every POST where a password or username keyword appears · exposing credentials on unencrypted channels.

◆ WHEN TO USE THIS

Auditing internal apps that still run over plain HTTP.
Pentests, awareness training, or breach investigations.

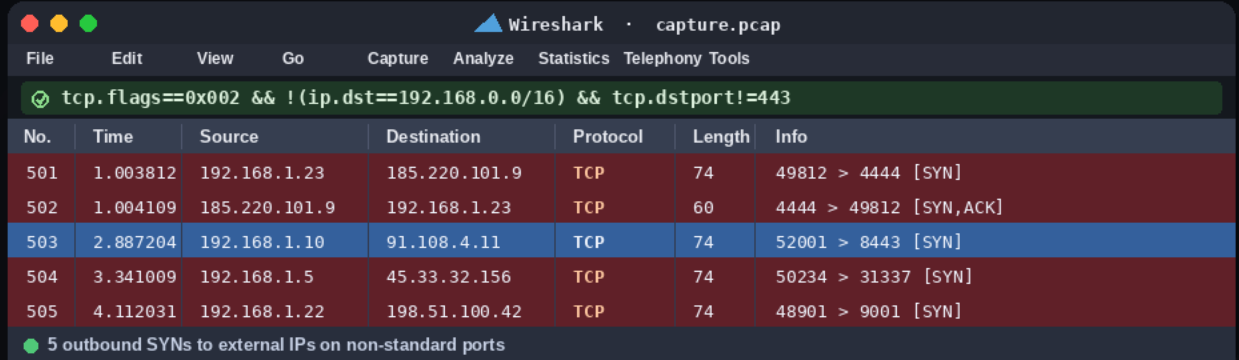
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X @keraattin

Suspicious Outbound

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Find connections leaving on non-standard ports.



Wireshark · capture.pcap

File Edit View Go Capture Analyze Statistics Telephony Tools

`tcp.flags==0x002 && !(ip.dst==192.168.0.0/16) && tcp.dstport!=443`

No.	Time	Source	Destination	Protocol	Length	Info
501	1.003812	192.168.1.23	185.220.101.9	TCP	74	49812 > 4444 [SYN]
502	1.004109	185.220.101.9	192.168.1.23	TCP	60	4444 > 49812 [SYN,ACK]
503	2.887204	192.168.1.10	91.108.4.11	TCP	74	52001 > 8443 [SYN]
504	3.341009	192.168.1.5	45.33.32.156	TCP	74	50234 > 31337 [SYN]
505	4.112031	192.168.1.22	198.51.100.42	TCP	74	48901 > 9001 [SYN]

5 outbound SYN's to external IPs on non-standard ports

`tcp.flags==0x002` ▶ Pure SYN · new outbound connection attempt

`!(ip.dst==192.168.0.0/16)` ▶ Destination is outside the local network

`!(ip.dst==10.0.0.0/8)` ▶ Also excludes the 10.x RFC1918 range

`tcp.dstport!=80/443` ▶ Not standard web ports · unusual traffic

Reverse shells and C2 beacons connect outbound on unusual ports.
This filter shows every new TCP connection leaving your network
to the internet on a non-standard port.

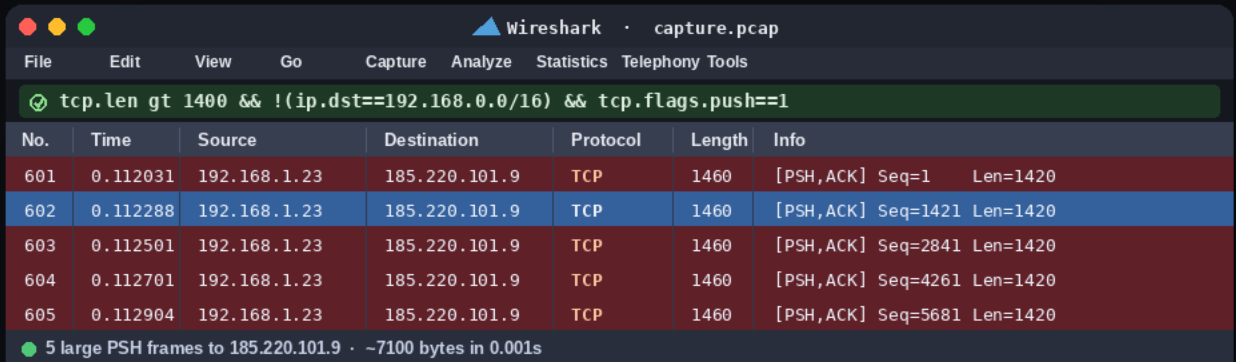
♦ WHEN TO USE THIS

After malware execution to find the callback channel.
Routine egress monitoring on critical workstations.

Data Exfiltration

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Catch large payloads being pushed externally.



Wireshark · capture.pcap

File Edit View Go Capture Analyze Statistics Telephony Tools

tcp.len gt 1400 && !(ip.dst==192.168.0.0/16) && tcp.flags.push==1

No.	Time	Source	Destination	Protocol	Length	Info
601	0.112031	192.168.1.23	185.220.101.9	TCP	1460	[PSH,ACK] Seq=1 Len=1420
602	0.112288	192.168.1.23	185.220.101.9	TCP	1460	[PSH,ACK] Seq=1421 Len=1420
603	0.112501	192.168.1.23	185.220.101.9	TCP	1460	[PSH,ACK] Seq=2841 Len=1420
604	0.112701	192.168.1.23	185.220.101.9	TCP	1460	[PSH,ACK] Seq=4261 Len=1420
605	0.112904	192.168.1.23	185.220.101.9	TCP	1460	[PSH,ACK] Seq=5681 Len=1420

5 large PSH frames to 185.220.101.9 · ~7100 bytes in 0.001s

tcp.len gt 1400 ▶ TCP payload larger than 1400 bytes

!(ip.dst==192.168.0.0/16) ▶ Destination is outside the local network

tcp.flags.push==1 ▶ PSH flag · sender is actively flushing buffer

&& ▶ All three together pinpoint bulk outbound sends

Exfiltration sends large rapid bursts to an external IP.

PSH flag means the sender is actively flushing the buffer.

Large payload plus external destination equals your alert.

♦ WHEN TO USE THIS

When DLP alerts fire or a host hits unusual bandwidth.

Investigating data theft suspects in incident response.

Follow TCP/UDP Stream

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Read a full network conversation in plain text.

```
Follow TCP Stream

> CLIENT · 192.168.1.5
POST /admin HTTP/1.1
Host: intranet.corp
User-Agent: curl/7.88.1
Authorization: Basic
  YWRtaW46cGFzc3dvcmQx
Content-Type: x-www-form
Content-Length: 32

user=admin&pass=hunter2

< SERVER · intranet.corp
HTTP/1.1 200 OK
Server: nginx/1.21.6
Date: Tue, 14 May 2026
Content-Type: text/html
Set-Cookie: session=
  abc123def456; HttpOnly
Content-Length: 4821

<!DOCTYPE html>...
```

Red bytes were sent by the client. Blue bytes by the server.

HOW TO USE

- 1 Right-click any packet in the session
- 2 Choose Follow > TCP Stream
- 3 A window opens with the full conversation

USE CASES

- Extract Base64 credentials from auth headers
- Reconstruct full HTTP, FTP and Telnet sessions
- Save stream to file for deeper offline analysis

♦ SHORTCUT

Ctrl + Alt + Shift + T · Follow TCP Stream

Color Rules

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Make anomalies impossible to miss at a glance.



TCP

Standard connections



DNS

Name resolution



HTTP

Unencrypted web



TLS

Encrypted sessions



UDP

Stateless transport



ERROR

Retransmissions / RST



ICMP

Ping and unreachable



ARP

Address resolution

ADD CUSTOM RULE



bash

View > Coloring Rules > +

Name : Suspicious Outbound

Filter : tcp.flags==0x002 && !ip.dst==192.168/16

BG : Red **FG** : White

♦ PRO TIPS

Rules evaluate top-to-bottom · order matters.

Export rules as files to share with the team.

Export Objects

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Extract files transferred over the network from a PCAP.

ACCESS PATH

```
File > Export Objects > HTTP
File > Export Objects > SMB
File > Export Objects > FTP-DATA
```

EXTRACT BY PROTOCOL



Images, scripts, HTML

HTTP



Windows file shares

SMB



Plaintext transfers

FTP



Firmware and PXE images

TFTP



Medical imaging data

DICOM



Save All in one click

ALL

TIP

Hash extracted files before analysis.
Save All extracts every object at once.

Statistics Menu

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Understand your capture without reading every packet.

Protocol Hierarchy

Statistics > Protocol Hierarchy

Every protocol with % of total traffic



Conversations

Statistics > Conversations

Host pairs sorted by bytes or packets



Endpoints

Statistics > Endpoints

Unique IPs and MACs with counts



I/O Graph

Statistics > I/O Graph

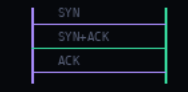
Traffic volume plotted over time



Flow Graph

Statistics > Flow Graph

TCP handshake sequence diagram



tshark

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Wireshark on the command line · scriptable and powerful.

CLI

Same engine

SCRIPTS

Pipe and grep

HEADLESS

Runs on servers

ESSENTIAL COMMANDS

```
# List network interfaces
tshark -D
# Capture live on eth0
tshark -i eth0
# Capture with a display filter
tshark -i eth0 -Y 'http.request'
# Read a PCAP file
tshark -r capture.pcap -Y 'dns'
# Extract specific fields
tshark -r cap.pcap -T fields \
  -e ip.src -e dns.qry.name
# Save filtered packets
tshark -r in.pcap -Y 'tcp.port==443' \
  -w out.pcap
```

♦ WORKFLOW

Capture with tshark on a remote server,
pull the PCAP, analyze with the GUI locally.

Profiles

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Save your setup · switch context in one click.

Default

Pentest

IR

Web Debug

DNS

↓ switch profiles from the bottom-right of the Wireshark UI

WHAT A PROFILE SAVES

- Color rules tuned for your use case
- Column layout · order, width and visibility
- Saved display filters and quick-access buttons
- Protocol preferences and decode-as settings

PROFILES TO BUILD

Pentest Cred hunt, C2 and exfil filter buttons

IR Red anomaly colors · delta time column

Web Debug Method, host, URI, response columns

DNS Query name, type, response, TTL columns

♦ SHARING PROFILES

Folders in %APPDATA%\Wireshark\profiles on Windows.
Copy the folder to share your setup with the team.

Cheat Sheet

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Save this slide · everything you need at a glance.

FILTERS

NOISE `!(arp || icmp || ssdp || mdns || llmnr || nbns)`

SCAN `tcp.flags.syn==1 && tcp.flags.ack==0 && tcp.window_size le 1024`

HOST `ip.addr==X.X.X.X && (tcp || udp) && !dns`

CRED `http.request.method=="POST" && frame contains "pass"`

DNS C2 `dns.flags.response==0 && dns.qry.name && frame.len gt 80`

OUTBOUND `tcp.flags==0x002 && !(ip.dst==192.168.0.0/16) && tcp.dstport!=443`

EXFIL `tcp.len gt 1400 && !(ip.dst==192.168.0.0/16) && tcp.flags.push==1`

SHORTCUTS

Ctrl+E Start / Stop capture

Ctrl+/, Focus filter bar

Ctrl+Alt+T Follow TCP stream

Ctrl+G Go to packet

Ctrl+T Set time reference

Ctrl+Down Next selected packet

Ctrl+R Open capture file

Ctrl+F Find packet

Ctrl+Shift+I Protocol Hierarchy

Space Next packet

Ctrl+M Mark / unmark packet

Ctrl+Shift+E Export packets

—◆
CONGRATS

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You're now Wireshark- literate.

Most people open Wireshark
and get lost. **You filter.**

- ◆ Save this for your next PCAP session
- ◆ Share with a teammate who needs this



LIKE



COMMENT



SHARE



SAVE



Instagram
@kerem.tech



X
@keraattin



YouTube
Kerem Tech



GitHub
keraattin